

RESEARCH ARTICLE

AUTOMATED AISI 317L SS THIN SECTION IMAGE POROSITY RECOGNITION AND SEMI-AUTOMATED PRIORITIZATION FOR PREDICTING THE YIELDING POSITION BY MEANS OF IMAGE PROCESSING

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ABSTRACT

Porosity is a common phenomenon in fabricated parts, which cause stress concentration and leads to yielding, brittle fraction and fatigue of structures which makes porosity recognition and analyzing very important to increase efficiency and decrease the defects of manufactured parts. Many researches have been done in order to detecting and recognizing pores of material by different numerical and experimental methods. In this paper, an algorithm is developed in order to detect the porosity of thin section images automatically and prioritize them for a semi-automatically prediction of yielding position. The algorithm results is the following parameters in a porous material: number of pores, position and size of pores, maximum and minimum distances of pores both from each other and from boundaries. The algorithm shows an accuracy of 83% in the prediction of the results for simulated finite element method (FEM) tensile tests, which makes the algorithm creditable to be used as a non-destructive testing method.

KEYWORDS

Porosity, Thin-section image, Image analysis, Finite element method, Aisi 317L

1. INTRODUCTION

Porosity or void fraction is the ratio or percentage of pore space volume to the total volume of the material between 0% and 100% (Lyons et al., 1996). In some industries, pores appear beneficial such as medical implant, aerospace and other light-weight structures, even porosity of reservoir rocks underground which specifies the amount of petroleum that a rock can store (Wang et al., 2013; Guo et al., 2019). However, in manufacturing and producing parts and structures, pores are not desirable since they have negative effects on strength which limits their applications. Despite the vast developments and adoptions, porosity appears because of incorrect processing parameters or building conditions. High temperature of the process is another reason of porosity (du Plessis et al., 2018). Absorption of nitrogen, oxygen, and hydrogen in the molten weld line of welding process are examples of porosity appearance in such processes. Effects of temperature, gravity, capillary and absence of mechanical pressure in the selective laser melting (SLM) method of additive manufacturing are other reasons for the appearance of porosity in materials (Ziolkowski et al., 2014).

Porosity characterization such as detection, recognition, and analysis including size calculation and prioritization is critical for a broad understanding of how porosity affects the mechanical and thermal properties of materials. Due to the appearance of this phenomenon, process qualification such as testing the produced parts has a highlighted role for decreasing the defects and increasing the quality of manufactured parts and structures (Malcolm et al., 2007). However, to our knowledge, an automated comprehensive method for analyzing the porosity of

fabricated parts has not been introduced yet. This is why the automated porosity detections and analyzing are interesting subjects for researchers especially in the additive manufacturing process. There are two testing methods for manufactured parts; destructive and non-destructive. Destructive methods such as the metallographic cross-section of parts in laboratories are expensive and time-consuming which may lead to higher error rates (Ziolkowski et al., 2014). In last decade, many advances are achieved in non-destructive methods such as the application of image analysis, intelligent systems, and pattern recognition techniques. Their precise results, multiple and rapid measurements and cost-effectiveness, made them popular (Ghiassi-Freez et al., 2012).

There are various researches on non-destructive methods of porosity analyzing. Malcolm and colleagues presented developments on the refinement of segmentation in image processing and highlighting the results by means of the specialized toolbox for metal foams (Malcolm et al., 2007). Modification of segmentation algorithm and denoising of CT slice images have not been considered in this paper. A group researchers investigated different pattern recognition methods for detection and analyzing pores of thin-section images of rocks (Richa et al., 2006). They conclude that estimation of porosity mostly depends on pattern recognition and image analysis techniques. A group researchers investigated the porosity of 316L SS specimens produced by additive manufacturing selective laser melting using X-ray computed tomography scan (Yusuf et al., 2017). The porosity of fabricated parts are attributes of the local melting of powder layers, rapid heating and cooling during the process. In other study, researchers used backscattered electron images of different types of stones for collecting data of the porosity distribution

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(Buckman et al., 2017). They presented the results with 2D contour plots of Matlab. With this method, the characteristics and distribution of pores in sandstones could be investigated and the experimental tests are not required.

A. Rabbani developed a new method of analyzing the connectivity and permeability of pores network in a specimen, which could be applied on 2D images to extract the pore and throat size distribution (Rabbani et al., 2014). Pores and throats assumed as spheres and cylinders, respectively in order to generate a pore network model. J. Ghiasi-Freez presented an algorithm for semi-automatic recognition and classification of porosity within thin-section images (Ghiasi-Freez et al., 2012). They extracted segmented images in order to use as an input of linear and non-linear classifiers. The non-linear process presented more accurate results than linear one. Recognition and analyzing the porosity of fabricated parts and structures have a critical role in different industries. They are applied to prevent accidents originating from yielding and the fraction of structures, which lead to both human and financial losses. The purpose of this paper is to develop an automatic porosity recognition and semi-automatic prioritization algorithm in order to detect pores of manufactured specimens.

The algorithm prioritizes pores in order of the most susceptible pores that lead to yielding the samples by a tensile force. For this purpose, a thin-section porous 317L stainless steel specimen is designed and meshed by Inventor software. Then the section images of the sample are imported to Matlab software. In the presented work, image enhancement algorithms have not been used since the quality of images were high. However, there are different methods for denoising and enhancement of the quality of the imported images, which may be used when there is a need (Ghiasi-Freez et al., 2012). Image segmentation is the next step that is the process of identification, recognition and highlighting specific subjects in an image (Ghiasi-Freez et al., 2012). This step automatically detects pores and semi-automatically prioritizes them by their positions and sizes. Finally, the results are validated by simulating the tensile test of the specimens using FEM. Figure 1 shows the workflow of the proposed method. In the following section, thin section images of porous fabricated part have been exported and properties of Aisi 317L SS have been specified.

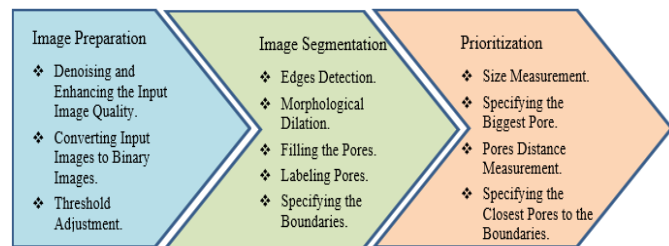


Figure 1: Workflow of the proposed algorithm to detect and prioritize pores of thin-section 317L S

2. MATERIALS AND METHODS

Different thin-section porous Aisi 317L SS specimens by dimensions $10 \times 5 \times 0.1$ cm have been designed by FEM. Figure 2 shows a few typical images of the thin-section designed specimens.

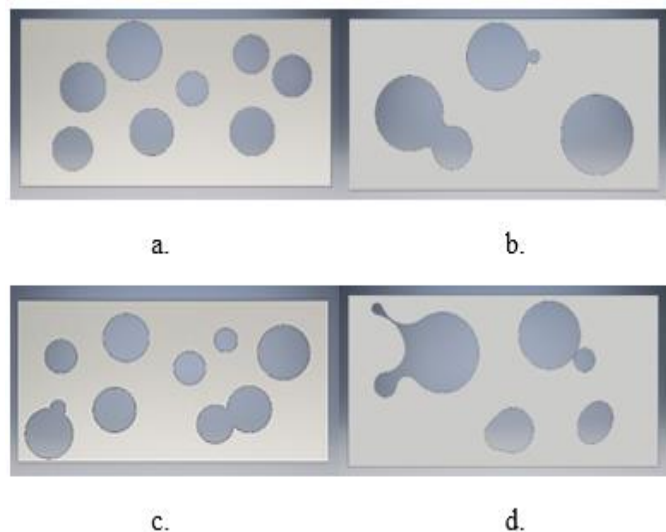


Figure 2: A few typical thin-section images of porous 317L SS specimen designed by FEM

317L SS is a low carbon austenitic stainless steel, which is corrosion resistance that makes it more affordable for chemical and petrochemical tanks and vessels. In addition, it has sensitization resistance during thermal processes especially welding with good formability and weldability which makes it more interesting for vast industries. The mass density of this type of stainless steel is $8 \text{ (g/cm}^3\text{)}$, yield strength is 317.16 MPa and Young modulus of 199.9 GPa. Table 1 shows the properties of Aisi 317L SS.

Table 1: Properties of Aisi 317L SS and their values	
Properties	Values
Density	$8 \text{ [g/cm}^3\text{]}$
Mass	1.51 [kg]
Yield Strength	317.16 [MPa]
Ultimate Tensile Strength	461.95 [MPa]
Young's Modulus	199.95 [MPa]
Poisson's Ratio	0.29
Shear Modulus	77.50 [GPa]

In the following section, the thin-section images of the porous 317L SS specimen have been imported into Matlab software. Then, the algorithm for automated recognition of porosity and prioritization of pores has been developed.

2.1 Research Method and Assumptions

Image analysis is a part of the computer-based process for the detection and quantification of considered subjects in an image (Martinez-Martinez et al., 2007). Diverse steps should be taken for analyzing an image, based on the goals and results. The first step of the work is image acquisition. In this part, section images of the samples have been extracted and taken into the next step. There are different ways for extracting the section images such as X-ray computed tomography (XCT) and FEM. In this paper, the thin-section specimens are designed in Inventor software by FEM, and images of the samples are exploited.

The section image of the specimen has been imported into the Matlab as shown in Figure 3(a). Next, the image has been converted to a binary image as shown in Figure 3(b). Thresholding factor considered 0.5 to create a binary mask containing the segmented pores of the specimen by exporting the edges. "strel" as a flat morphological structuring element algorithm which is essential for morphological dilation is applied (Matlab Users Guide, 2016). The output result is shown in Figure 3(c). Finally, the holes have been filled and edges of the specimen omitted. Figure 3(d) shows pores of the sample.

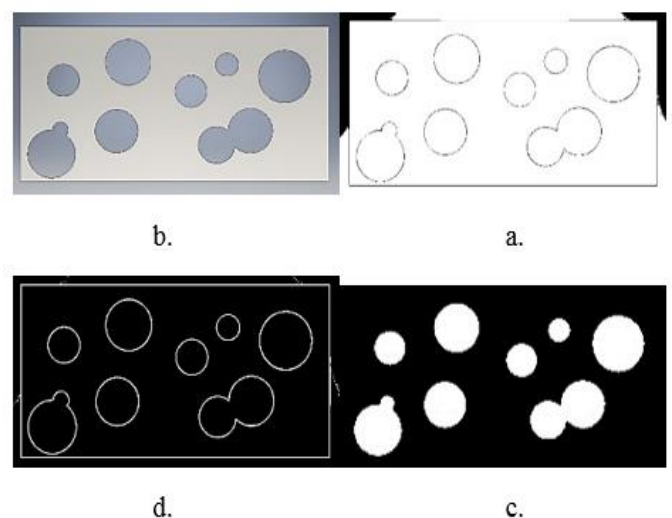
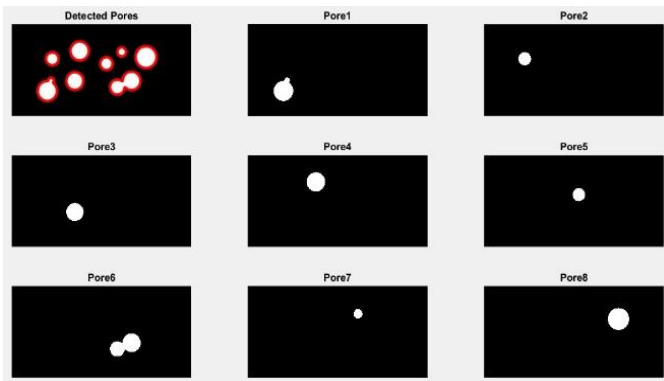


Figure 3: The automatic detection of pores in different steps: (a) Original image, (b) Binary image, (c) Image is dilated and edges are detected, (d) Holes are filled and edges of specimen are removed

In the next step, the filled holes are labeled and their boundaries are

specified with a red color and entitled automatically as shown in Figure 4:



In the next step, the identified pores will be prioritized based on their sizes and positions. For this purpose, centers of pores have been specified. We use "regionprops" algorithm that returns measurements for a specific property (Matlab Users Guide, 2016). For example, centroids of connected ingredients that create a region such as holes in a binary image. Then, the centroids are marked and titled as shown in Figure 5:

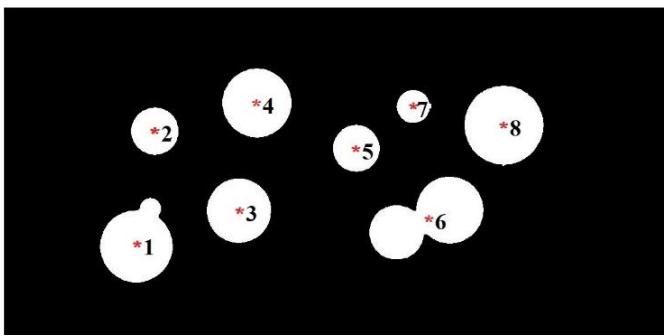


Figure 5: Pores' centroids of thin-section 317L SS sample specified and entitled

The maximum and minimum relative tensile strength limits of the real porous materials are shown in Eqs. (1) and (2), respectively (Haynes, 1971). They are correct if early failures do not happen due to accidental flaws.

$$\sigma_{Rel(max)}^0 = \frac{\sigma_{TS(max)}}{\sigma_{TS}^0} = 1 - \varepsilon, \quad (1)$$

$$\sigma_{Rel(min)} = \frac{\sigma_{TS(min)}}{\sigma_{TS}^0} = \frac{1 - \varepsilon}{K_p}, \quad (2)$$

$$\sigma_{TS(max)} = \sigma_{TS}^0 (1 - \varepsilon), \quad (3)$$

It is a where σ_{Rel}^0 is the relative tensile strength of pore-free material and σ_{Rel} is tensile strength of porous material, σ_{TS}^0 is tensile strength of porous material and σ_{TS} is the tensile strength of pore-free material. K_p is theoretical (elastic) stress-concentration factor of pores and it is equal to

$$K_p = \frac{1 - E_{Rel}}{\varepsilon} = \frac{1 - E_{Rel}}{1 - \rho_{Rel}}, \quad (4)$$

where E_{Rel} is relative elastic modulus and ρ_{Rel} is relative density. ε is fractional porosity and is equal to (Duta and Bose, 2012):

$$\varepsilon = 1 - \frac{\rho}{\rho^0}, \quad (5)$$

$$\rho = \frac{M}{V} \quad (6)$$

where ρ , M and V is density, mass and volume of porous material, respectively. Therefore, the size of pores and their distribution have a highlighted role in the magnitude of relative tensile strength of the material (Haynes, 1971; Firstor and Podrezov, 2001).

In addition, according to pore distance from the boundaries effects the

yielding position (Xu et al., 2011). Growing the pore size increases the stress zone. Further, stress concentration is more on pores that are closer to the surfaces. So, in order to prioritize the most susceptible pores that cause stress concentration and yielding of the specimen in a tensile test, the biggest pore and the nearest pores to the boundaries have been specified automatically. The default unit of the measurements is pixel, which has been converted to mm. The biggest pore is then figured and its size is displayed in mm^2 . Figure 6(a) shows the original image of a typical example. The biggest pore is detected and area calculated as shown in Figure 6(b). The algorithm calculated the size of the biggest pore in this section image as 405 mm^2 .



Figure 6: (a) Original image of a typical example, (b) The biggest available pore in the thin-section image

Using the information of the centroids of pores, it is feasible to measure the distances between pores and calculate the maximum and minimum distances. For this purpose, Euclidean distance between each pair of pores' centers have been measured. By comparing each distance with other ones in a loop, the maximum and the minimum distances are specified. The centroids for a typical section image are detected and shown in Figure 7(b). The distance between pores 1 & 3 and 1 & 2 are chosen and indicated as the maximum and the minimum distances between the image pores, respectively.

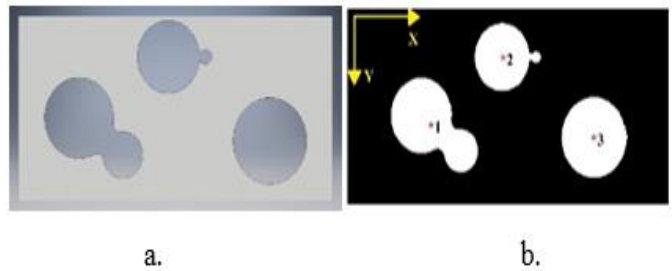


Figure 7: (a) Original image, (b) Centroids. Yellow arrows define the X and Y axis in the coordinate.

The minimum distance between pores in this typical example is 31 mm which is the distance between pores 1 and 2. Closest pores to the boundaries are candidates of the stress concentration where have the maximum and minimum values of i and j indicating the center coordinate of the pores in the image matrix in X and Y-axis, respectively. First, by using the location of the centroids, the i and j values of each pore on the X and Y-axis have been computed. Then, pores with maximum and minimum values of i and j are specified. The closest pores to the boundaries for a typical example, which is specified by 'C', are shown in Figure 8(b).

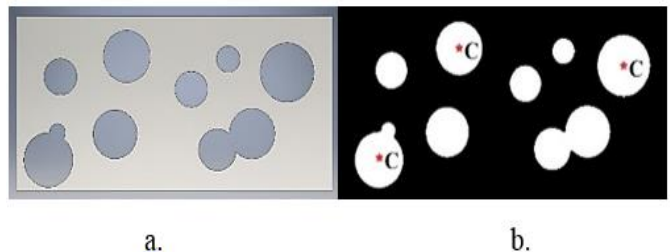


Figure 8: (a) Original image, (b) Closest pores to the boundaries.

3. RESULTS AND DISCUSSION

By analyzing the results achieved from the proposed algorithm, it is feasible to predict the stress concentration regions that lead to yielding and fraction. The algorithm considers the following parameters: number of pores, position and size of pores, maximum and minimum distances of pores both from each other and from boundaries. The results of the image

analysis for a typical example are shown in Figure 9:

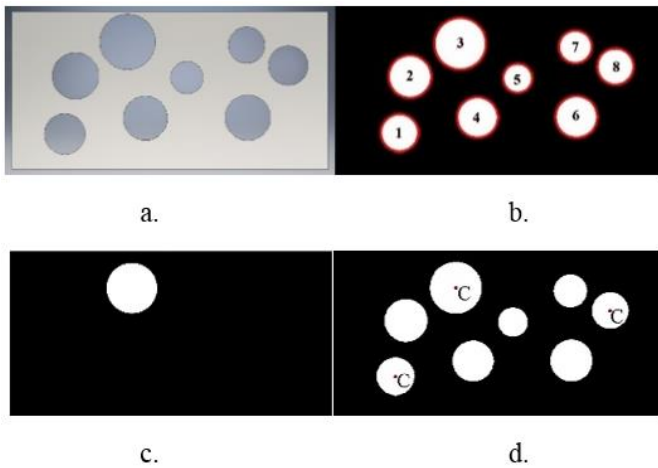


Figure 9: (a) Original image, (b) Detected pores, (c) Biggest pore, (d) Closest pores to the boundaries specified by 'C'

In this example, 8 pores have been detected and the minimum distance is 14 mm between pores 7 and 8. The pore 3 of the specimen is the biggest pore with the size of 207 [mm]^2 , which is one of the closest pores to the boundaries. By considering Eqs. (1) to (6), the fractional porosity has the highest value for the pore 3, which leads to reduction of the tensile strength. Further, because of the location near to the boundary, this pore is the most probable region for the stress concentration. So, by considering the algorithm results and Eqs. 1 to 6, the pore 3 is the most vulnerable region for fatigue to be started in tensile test. In order to investigate on the validity of the proposed prediction method for thin-section sample of Fig. 9, a tensile test by means of FEM have been used. The specimen is meshed according to the details provided in table 2. It is then taken into a tensile test of 1000 MPa in the direction shown in Figure 10 and 11 where the simulation results are shown with colored contours. The other side of the sample fixed by 50 kN.

Table 2: Mesh setting and values for Aisi 317L SS sample

Mesh Setting	Value
Avg. Element Size (fraction of model diameter)	0.05
Min. Element Size (fraction of avg. size)	0.05
Grading Factor	1.5
Max. Turn Angle	60 deg

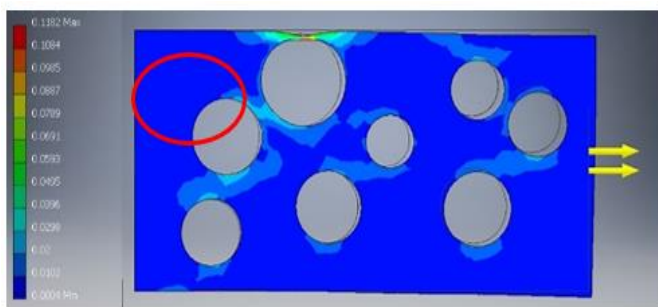


Figure 10: Equivalent strain of 317L SS porous sample in a tensile test

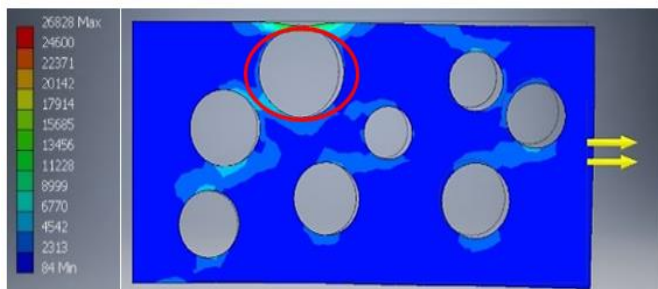


Figure 11: Von Mises stress [MPa] of 317L SS porous sample in a tensile test

The maximum and the minimum values of stress results, displacement,

and strains for the typical example of Figure 9 are shown in Table 3. The stress of 26828 MPa concentrated at the edges of the pore 3 in the simulated tensile test as shown in Figure 11 with a red circle. This is the same pore as it was chosen and predicted by the algorithm in Figure 9. The maximum of equivalent strain is 0.12 at the edge of the pore 3. So, this point is the most susceptible location for yielding and fracture of the specimen which was predicted by the proposed method and confirmed by FEM analysis.

Table 3: Results of simulating tensile test for porous Aisi 317L SS sample

Result	Min.	Max.
Von Misses Stress	84.43 [MPa]	26828.3 [MPa]
Stress xx	-4475.5 [MPa]	26478.9 [MPa]
Stress yy	-6077.49 [MPa]	3542.35 [MPa]
Stress zz	-750.31 [MPa]	972.19 [MPa]
Displacement	0 [mm]	1.98 [mm]
Equivalent Strain	0.0004	0.12
Strain xx	-0.019	0.13
Strain yy	-0.044	0.011
Strain zz	-0.038	0.009

In another validation, a thin-section porous Aisi 317L SS sample has been designed and pores are detected automatically as shown in Figure 12. In this example, there are 8 pores that have been detected automatically. The minimum distance is measured 12 mm which shows the distance between pores 5 and 7. The pore 6 is detected as the biggest pore of the specimen with the size of 158 mm^2 .

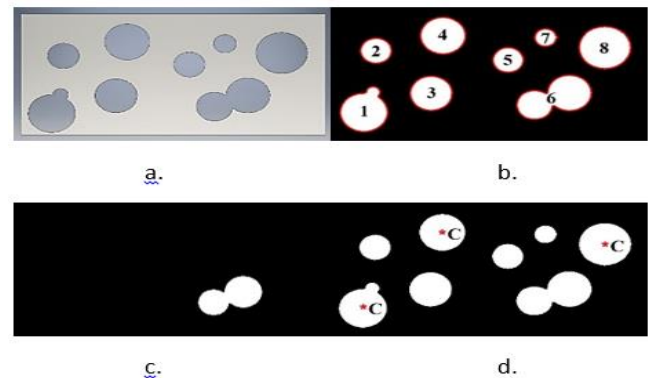


Figure 12: (a) Original image, (b) Detected pores, (c) Biggest pore, (d) Closest pores to the boundaries specified by 'C'.

Although the biggest pore is the pore 6, the nearest pore to the boundaries is the pore 1. On the other hand, size of the pore 1 with 119 mm^2 is also comparable with the size of the pore 6. By considering the direction of the force in the tensile test and the distance from the boundaries as shown in Figure 12(d), the pore 1 has more important role than the pore 6. The pore 1 increases the value of fractional porosity leading to the reduction of the tensile strength. By considering Eqs. (1) and (2), this pore is predicted to have higher stress concentration than other pores. To validate the prioritization of the algorithm, the model of the sample is imported to the simulation section of Inventor software. 1000 MPa tensile force is applied on the sample by FEM. The results are shown in table 4, Figure 13 and 14. Both Von Misses Stress and strain are concentrated on the edge of pore 1. Yielding starts from this point as shown in Figure 13 and 14. The pore 1 has higher value of fractional porosity and lower relative tensile strength.

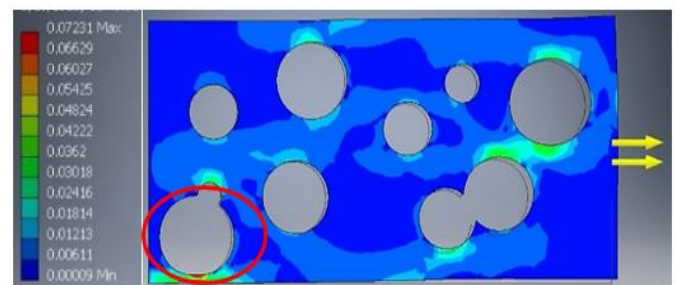


Figure 13: Equivalent strain of 317L SS porous sample in a tensile test

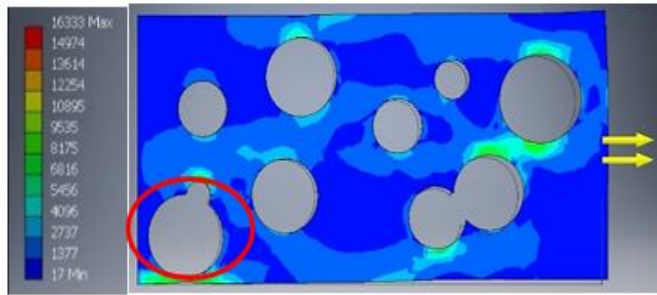


Figure 14: Von Misses stress [MPa] of 317L SS porous sample in a tensile test.

Table 4: Results of simulating tensile test for porous Aisi 317L SS sample

Result	Min.	Max.
Von Misses Stress	17.47 [MPa]	16333.3 [MPa]
Stress xx	-1709.03 [MPa]	16646.4 [MPa]
Stress yy	-3549.03 [MPa]	3793.84 [MPa]
Stress zz	-556.20 [MPa]	2267.86 [MPa]
Displacement	0 [mm]	1.64 [mm]
Equivalent Strain	.0001	0.0723
Strain xx	-0.0077	0.0821
Strain yy	-0.0238	0.0175
Strain zz	-0.0247	0.0063

It is noteworthy to mention the proposed algorithm expects a higher strain and a higher Von Misses Stress in the neighborhood region between pores 5 and 7 since the minimum distance is detected between the pores 5 and 7. However, FEM results show the higher values in the region between the pores 6 and 8 than pores 5 and 7. This is inconsistency with the prediction of the proposed algorithm. The reason for this erroneous prediction originates from the technique that is used in the proposed algorithm to measure the minimum distance. The algorithm measures the distance between centroids without considering the size of pores. If we change the distance criterion from centroid-to-centroid distance to edge-to-edge distance, the minimum distance will be between pores 6 and 8 where the FEM results show higher values. Therefore, the prediction will be compatible with the FEM.

4. CONCLUSION

In this paper, a non-destructive image processing-based method has been developed that automatically detects and recognizes porosity. It also prioritizes pores of thin-section Aisi 317L SS specimen semi-automatically, in order to predict the stress concentration position, which leads to yielding of the sample from that point. For validation of the algorithm, 23 images of thin-section 317L SS sample have been analyzed and compared with FEM results. The proposed method shows the accuracy of 83% in finding the porosity of the thin-section sample and location of the stress concentration. However, we showed using more advanced distance measurement technique can increase the accuracy of the proposed method. The algorithm considers the following parameters: number of pores, position and size of pores, maximum and minimum distances of pores both from each other and from boundaries. The presented results show that the method uses a simple and fast algorithm to recognize and prioritize porosities. It detects the porosities automatically and prioritize them semi-automatically in order to predict yielding position in the tensile test. Both the pore size and the distance to the boundaries play an important role in the proposed method. The ongoing race towards smaller distance (edge-to-edge or edge-to-boundaries) and bigger pore size in the prediction of the yield position therefore both speaks in favor for this technique. Some aspects of future work would include developing an algorithm to contain more color range of material for detecting pores and developing an algorithm to contain compressing test. Also, this method has a high potential to be developed by artificial intelligent as a full automatic pore detection and yielding region prediction algorithm.

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